



NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI – 15
DEPARTMENT OF MECHANICAL ENGINEERING

Cycle Test-1

Programme and specialization: B.Tech. - Instrumentation and Control Engineering
Course code/Title : MEIR11/Basics of Mechanical Engineering
Max. Marks: 20

Date: 14.10.2025

Time: 3.45 pm – 4.45 pm.

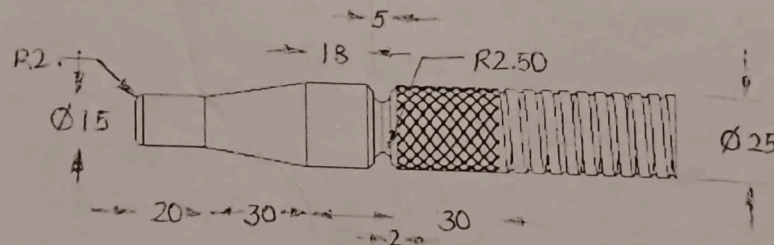
Answer all the questions.

Part-A

1. Explain the principle of operation of IC engine using diesel in which one cycle of operations is completed in two revolutions of the crankshaft. [4 M]
2. What is meant by scavenging in two-stroke engine? [1 M]
3. What are the functions of the fuel injection pump in a Diesel engine? [1 M]
4. Calculate the number of sparks per minute produced by the spark plugs of a four-stroke four-cylinder petrol engine running at 3000 rpm? [1 M]

Part-B

1. Explain the purpose of a split pattern in casting. Illustrate with a suitable example. [3 M]
2. Identify a suitable metal joining process used for electrical connections and describe its working principle. [3 M]
3. Suggest the most appropriate metal forming process for manufacturing a crankshaft and justify your selection. [2 M]
4. For final job shown below (assume all dimensions are in mm), determine the optimal sequence of operations required on a given workpiece, on a Lathe. [2 M]



5. A stainless steel rod of 15 mm diameter and 200 mm length is to be reduced to 12 mm diameter by turning on a Lathe in one pass. The spindle speed is 650 rpm, and the tool feed rate is 180 mm/min.
 - (a) Calculate the cutting speed (m/min)
 - (b) Determine the material removal rate (MRR)
 - (c) Estimate the machining time required. [3 M]